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SOL-04_evidence_class

SOL- — Evidence-Class-at-Closure Enforcement (the Discriminator as a Seam)

Field	Value
Revision	1
Last modified	2026-07-22T21:05:00Z
Rank	2 (the highest *prevention* leverage — it mechanizes the one predictor that held in every corpus)
Closes	PC-3 (source-green read as done), PC-8 (wrong-layer oracles), PC-9 (with SOL-01 — evidence preconditions per status)
Seam	status-write (composes with SOL-01: the custody chain's evidence files are class-checked before the terminal write is permitted)
POC	`poc/sol04_evidence_class/` — GREEN 5/5, RED-first
Anchors mechanized (no new anchor)	§11.4.115(F) evidence-class clause, §11.4.108 layers, §11.4.123

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. The measured problem — and the discriminator

The single strongest empirical result across all four inputs:

> **Every fix whose done-claim was preceded by an on-target RED→GREEN polarity flip that held recorded ZERO reopens; every fix confirmed at source-green bounced.** (ROOT_CAUSE §7 — the positive control; re-verified against the live ledger; no counter-case found in any corpus.)

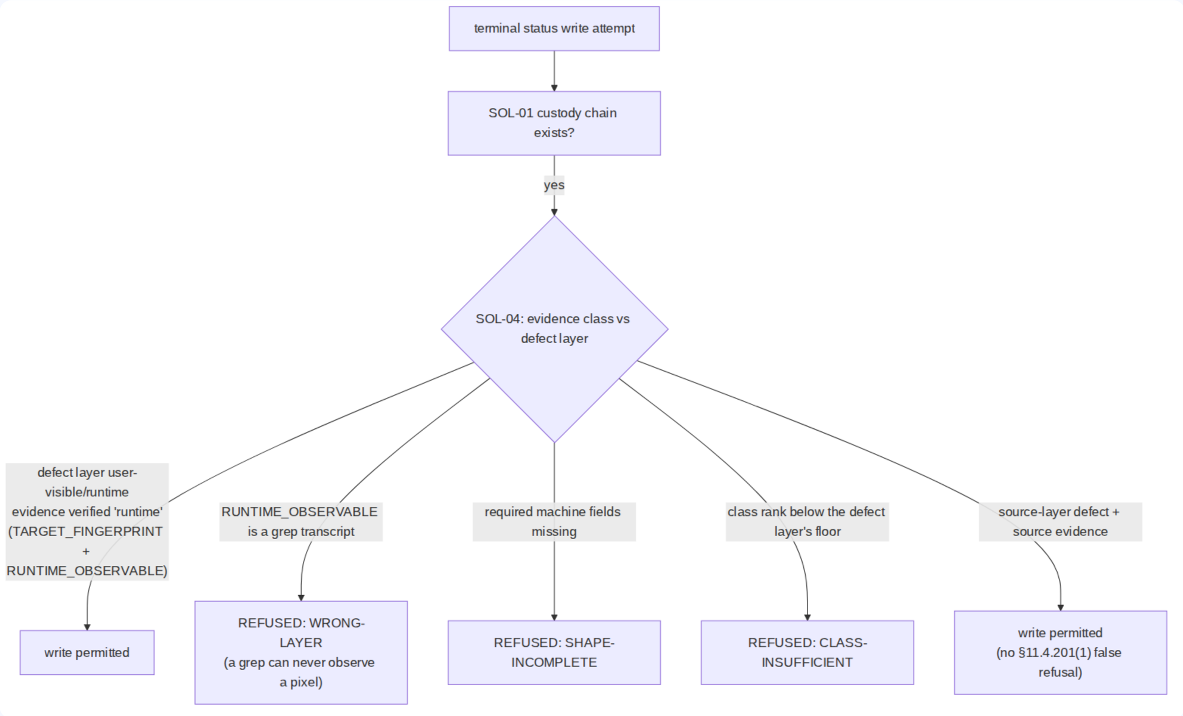
Cross-corpus confirmation:

Corpus	Closure evidence class	Outcome
Corpus 1 (ATMO-Sphere)	historically source-green	52% recorded reopen rate (under-count); 6/6 operator-sampled "done" items broken; 69 PENDING-BUILD claim-moves [M16]; QA entered with 1/25 validated past layer 1 [M14]
Corpus 2 (claude_toolkit)	hermetic sandbox (wrong layer)	"whole suite green while EVERY router alias on the real host was bricked" (its own forensic note); median 3, max 7 attempts per defect
Corpus 3 (helix_code root)	76% of fix commits cite a test, 46% runtime evidence, closures cite RED→GREEN + mutation	**median attempts-to-stick = 1; ~0.3% git-corrected reopen rate**
Corpus 3's ONLY bouncing family	i18n — original PASS was a unit test asserting the Noop echo (wrong-layer oracle)	the corpus's biggest fix (74 packages) AND its only silent reopen — the discriminator's predicted loser, n=1, sign right

External mechanism (R2.1/R2.2/R2.5, S18/S19/S27): fault-detection power lives in the ORACLE; execution metrics don't predict detection, checking metrics do; the oracle problem is the field's named central difficulty. PC-8's canonical instance: a pixel-layer defect "verified" by a file-existence check plus five greps.

. The mechanism

Make the evidence CLASS a refusable property of the closure, verified by machine fields — never by a label:



- **Closed sets.** Defect layers {user-visible, runtime, artifact, source}; evidence classes {runtime > artifact > source} with a rank floor per layer. Source evidence can close only a source-layer item; a user-visible defect requires runtime-class evidence.
- **Class is proven by machine fields**, not asserted: `runtime` requires `TARGET_FINGERPRINT` (read from the target, §11.4.115(F)) + `RUNTIME_OBSERVABLE` ; `artifact` requires `ARTIFACT_PATH` + `ARTIFACT_SHA256` ; `source` requires `SOURCE_REF` . A label without its fields is `SHAPE-INCOMPLETE` .
- **The anti-echo rule** kills the exact PC-8 shape: a `RUNTIME_OBSERVABLE` that *is* a grep transcript is refused as `WRONG-LAYER` — the wrong-layer oracle dies by construction, which is precisely what corpus 3's one bouncing family lacked.

- **Composition with SOL-01/SOL-02:** SOL-01 requires the chain to exist;
3 SOL-04 requires its evidence to be of the right class; SOL-02 requires it to exist *for the release candidate*. Three refusals, one custody pipeline.

. POC results

RED (exit 1, artifact missing) → GREEN **5/5**: real runtime evidence accepted; grep-transcript- wearing-a-runtime-label refused (`WRONG-LAYER`); fieldless label refused;
4 source-on-source accepted (negative control — legitimate source-layer work is not false-refused); unreadable evidence is `BLIND(2)` , never accepted.

. The failure this makes IMPOSSIBLE

A terminal status backed only by source-layer evidence **cannot be written** for a defect whose layer is runtime/user-visible — the 69 PENDING-BUILD claim-moves [M16], the pixel-defect-by-five- greps closure [PC-8], and the corpus-2 "sandbox-green, host-bricked" release pattern all cross this seam and are refused at it.
5 PC-9's ambiguity dissolves as a side effect: each status's evidence *precondition* is now a checked property, not a vocabulary argument.

. What it still does NOT catch (honest boundary)

1. **Fabricated runtime fields.** A writer can invent `TARGET_FINGERPRINT` . The checker proves shape and layer, not provenance; provenance is the §11.4.115(F) harness-written-verdict rule (verdict files written by the harness, fingerprint read from the target at run time) + SOL-02's candidate-fingerprint join, where an invented fingerprint fails to match the real candidate and reads as absence. Layered.
2. **Oracle calibration.** Runtime-class evidence from a mis-calibrated analyzer still passes; §11.4.107(10) goldens remain necessary-not-sufficient (stated boundary carried from §11.4.146(D3)).

3. **The layer taxonomy is authored data.** Mislabelling a defect as `source`-layer to lower the floor is possible; countermeasure is review of the layer field (cheap to spot: the item's own description names user symptoms) — detective, not preventive. UNPROVEN: an automatic layer classifier from item text was not attempted.
4. **Echo-detection is a closed pattern list** (`grep:` / `source-grep` prefixes in the POC); a determined writer can phrase an echo differently. The negative pressure comes from the §1.1 mutation pairing (a guard whose evidence cannot fail its golden-bad is refused there), not from string-matching alone.